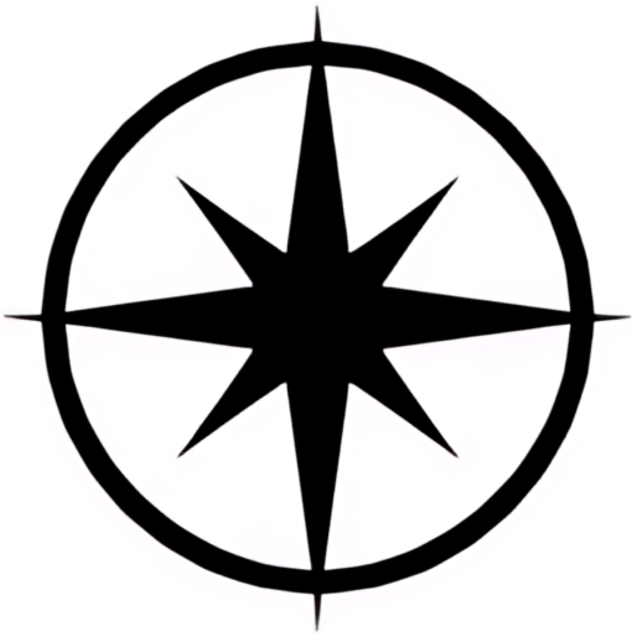

1)

$$\frac{n(3n-1)}{2} \qquad \frac{K(3K-1)}{2}$$

$$\frac{K+1(3K+2)}{2} = \frac{3K^2+2K+3K+2}{2}$$

$$3K+3-2 = 3K+1$$

$$\frac{3K^2-K}{2} + 3K+1$$

$$\frac{3K^2-K+6K+2}{2}$$

$$\frac{3K^2+5K+2}{2}$$

$$= \frac{(K+1)(3K+2)}{2}$$

if n^2+5 is odd then n is even

$$n^2+5=k+1$$

$$n=2k$$

$$2k+5=k+1$$

$$2k-k=1-5$$

$$k=-4$$

$$n=2\cdot(-4)=-8 \text{ even } \checkmark$$